TO: Medical Director, Medicare Shared Savings Program ACO

FROM: Vansh Chanchlani & Sagarika Raju, ISE 503 Health Analytics, University of Southern California

RE: Deploying a Predictive Risk-Stratification Tool to Identify High-Cost Medicare Patients Before Costs Are Incurred

---

We recommend that you incorporate a claims-based predictive risk-stratification model into your annual care management cycle beginning in the upcoming plan year. Using only prior-year Medicare claims, our model identifies future top 5% spenders with an AUC-ROC of 0.878 and flagging your top 10% by predicted risk score captures 72% of next year's high-cost patients, who account for 56% of all projected Medicare spending. Acting on this ranked list each January allows your care coordinators to intervene before avoidable hospitalizations occur.

Approximately 5% of Medicare members account for 50% of overall spending, yet high-cost patients are only discovered after it is too late to act. Our goal was to create a forward-looking priority list for your care management team by forecasting the top 5% expenditure in Year 2 using just Year 1 claims. Fee-for-service Medicare enrollees with 12 months of Part A and Part B membership are the target group; Medicare Advantage is not included because of incomplete claims. Your nurses and coordinators may assign limited outreach capacity to precisely the correct patients thanks to the use case, which is a graded risk-score list created every January. The tool's scope is limited to identification; it does not take the role of clinical judgment.

We used the CMS Synthetic Medicare Public Use Files, which are a certified government source that mirrors real Medicare Research Identifiable Files that link 2022 beneficiary records to inpatient, outpatient, and carrier claims for 3,339 patients. We created 30 features that included past spending, hospitalization type, readmissions, diagnosis burden across eight chronic disease categories, and year-over-year spending trends. XGBoost exceeded our Logistic Regression baseline (AUC = 0.878 versus 0.828). Prior total spending, physician payment intensity, and spending growth rate were identified as dominant predictors in SHAP analysis, which is consistent with Chechulin et al. (2014) and Rose (2016) findings. Primary limitation: results should be prospectively checked on your entire panel before deployment.

We recommend three steps: first, run the model annually on your enrolled panel using the previous year's claims; second, flag the top 10% of predicted-risk patients for proactive outreach on a 5,000-member panel (500 targeted contacts rather than 250); and third, pair each flag with its SHAP explanation so coordinators can tailor the intervention to the specific clinical driver. Preventing even 15% of needless hospitalizations among flagged patients would result in a return on investment that much outweighed program expenses, and for your most vulnerable members, earlier intervention is not only financially beneficial, but also clinically significant.